

The Internet of Energy

From smart products to smarter systems



Added transparency, enhanced productivity and advanced intelligence

Digitalization is an opportunity and an enabler. It empowers industries to tackle their current challenges: for example, implementing improvements in the areas of performance, health, safety, and environment as well as reducing costs and risks. However, to benefit from digitalization, it's first necessary to generate data. While the benefits are already viewed as common sense in many industries, the transmission and distribution industry wasn't able to profit from this industrial revolution for a long time. By merging traditional physics and information technology, we're making a fundamental shift in the industry. With smart sensors, cloud computing, and IoT, the limitations of the past are now making way for intelligent products and systems. The future is about scaling up product intelligence by combining theoretical knowledge, operational data, and digital twin data to challenge the physical frontiers of products and systems. The cloud will be the default platform for data storage and complex asset analytics that offers management recommendations on demand and in real time. The goal is to shape the transmission grids of tomorrow using smart products and smarter systems.

Welcome to the Internet of Energy!

We shifted an industry paradigm in 2018 with the introduction of Sensformer®. Since that time, all our transmission products have connectivity features built-in by default. With Sensformer and Sensgear®, the high-quality substation products from Siemens Energy are now elevated to a new digitalized level. Just like Sensformer, the connectivity of all products is ensured with a smart and robust IoT gateway that securely transmits a minimum amount of required information to a cloud-based storage and visualization platform.

By adding more data like weather information, even more potential emerges: Is there a storm coming? Is the sun heating up the asset? Will the wind get stronger, and so wind energy input will increase? Now this revolutionary connectivity concept that was initially introduced for transformers is the first to conquer the entire transmission products industry.



Products are connected with a smart and robust IoT gateway to a cloud-based storage and visualization platform

Sensproducts® – The next level for transmission grids

- System intelligence: combination of Sensformer® and Sensgear® for improved substation management
- Digital twin technology: controlled overload of a complete network
- Cloud visualization: dashboard with a full substation overview
- Status data: real-time data overview available everywhere
- Custom support concept
- Latest data security standards

From “connected” products to systems

Substations continuously generate and remotely transfer data to a control room using a standard multi-layer architecture. These traditional SCADA systems also assess the overall status of the assets and make decisions about their short-term performance or replacement.

The most attractive and demanding challenge is to combine the analog electrical transmission with a digital system for increase value creation and get more insights into the operational life of the assets. With connected products, you can augment the traditional SCADA architecture (as shown in the figure on the next page). All assets in a high-voltage substation can transmit data to a highly secure cloud-based platform that offers a parallel information channel to the SCADA system in real time. Data is analyzed by complex algorithms on this platform and as a result, the conventional control and protection philosophy can now be combined with a more agile information channel into a cloud platform for faster analytics.

From products to system intelligence

One major advantage of a cloud computing-based approach is the rapid interpretation of the status data from a system using a combination of products that's much faster than legacy systems.

More complex data management can also be implemented by using intelligent PLCs or IEDs than in typical SCADA systems. The analytics are more accurate, computations are faster and more reliable, and response times are lower. The breakthrough benefit of this innovative approach is the opportunity to integrate operational information in a dynamic way.

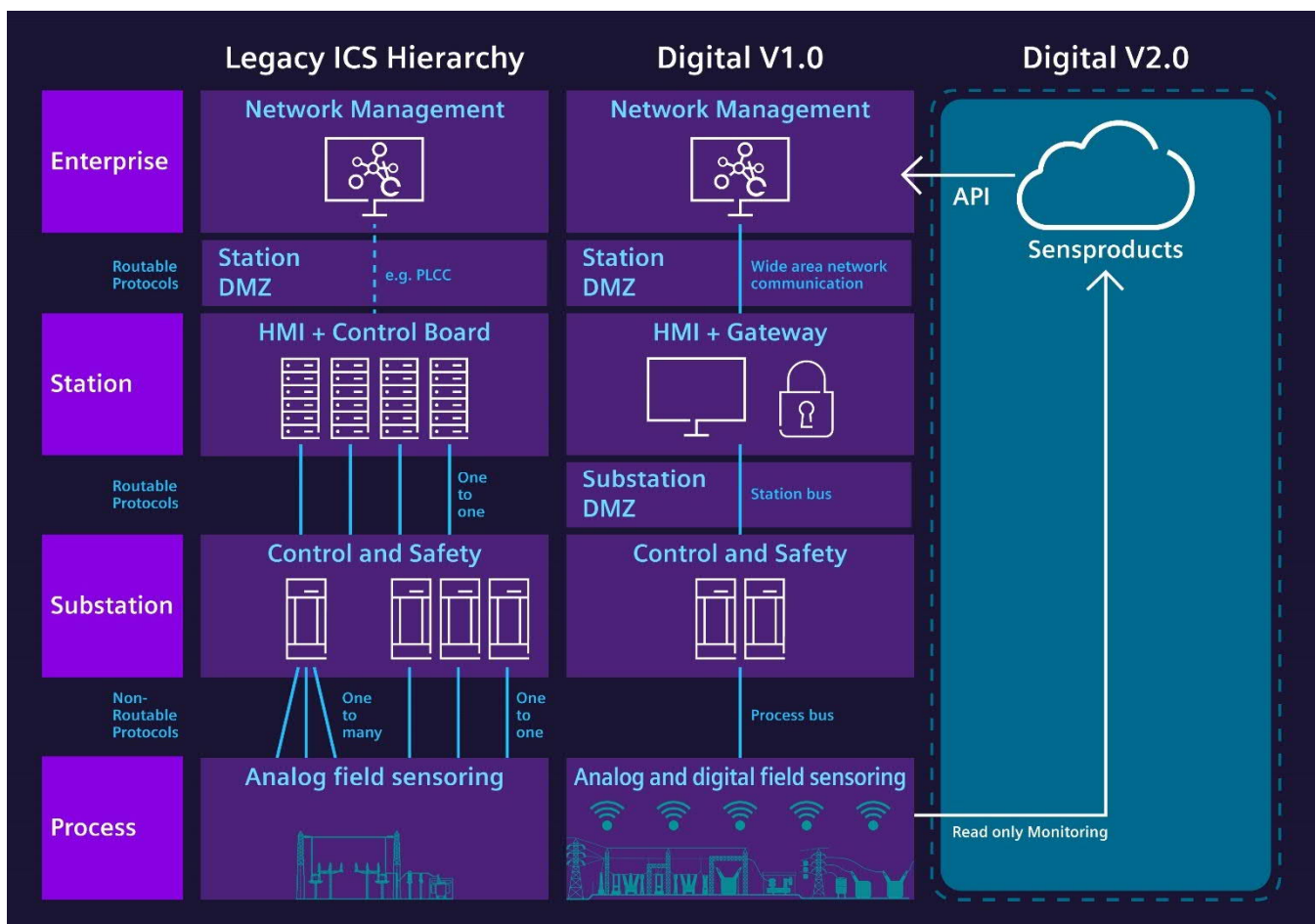
Faster data analytics leads to enhanced substation systems reliability and increased workforce productivity using easy to navigate apps that offer important asset or system information in near real-time, anywhere, on demand, and with clear management instructions. Fault localization and management and even the prevention of fault occurrence is now easy with connected products and systems. Dynamic systems overload is now achievable in a controlled manner without damaging products when Sensformer and Sensgear Advanced are combined with their digital twins.

From diagnostics to predictive and prescriptive Intelligence

With more real-time operational data provided by the assets, it's possible to develop very complex algorithms using artificial intelligence and machine learning that extract valuable information, maximize equipment utilization, and compute its residual life. Cloud connectivity easily allows all data to be collected from different sources for faster analysis in an integrated and more reliable system – moving away from condition-based monitoring toward continuous monitoring.

Convert your substations into data hubs

Just imagine: With all your connected products giving you a lot more information about your operational assets, you're now able to better manage your assets, make informed decisions, think about correlating events by connecting the dots (for example, the last operational date of your circuit breaker), identify and measure creepage currents, identify weak links in the system, optimize asset operations with an increased analysis of ambient weather conditions, and much more. You can customize your dashboards and monitor your assets based on all your specific needs and requirements.



Cloud-based platforms represent a parallel information channel to classical multi layer architectures, in real time

Optimize your CAPEX and OPEX

Advanced data analytics will increase the efficiency and sustainability of assets and operations and thereby reduce your CAPEX and OPEX. The system intelligence generated from the combination of connected products provides an additional basis for overcoming the limitations and complexity of traditional substations and addresses the simple monitoring needs of system operators, without replacing the current control and protection philosophy – and it simultaneously serves as a valuable source of information.

Join the journey...

Digitalization happens at a speed that's unprecedented in the transmission product industry. During the Sensformer journey that began in 2018, the solution was already evolving through the scaling-up of product features. The new applications and benefits provided by Sensproducts® and their digital twins are numerous for transforming traditional substation operations. The increasing and ever-evolving network complexity raises the bar for transmission system operators in terms of guaranteeing a high and safe level of power continuity and maximizing the productivity of their systems. We're confident that our Sensproducts will support them in achieving or redefining their operational goals.

Data management and security

Siemens Energy fully understands the ever-increasing cybersecurity requirements. As one of the main founders of the Charter of Trust, we ensure that we comply with state-of-the-art security and encryption technologies according to ISO/IEC 27001. An end-to-end encryption method is used for data transmission to cloud storage. Each Sensformer or Sensgear has a unique ID that's also used for encryption. The transmission takes place via HTTPS with 256-bit TLS encryption. We also comply with best-in-class data handling and management guidelines to prevent data breaches and ensure high security in the cloud.



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